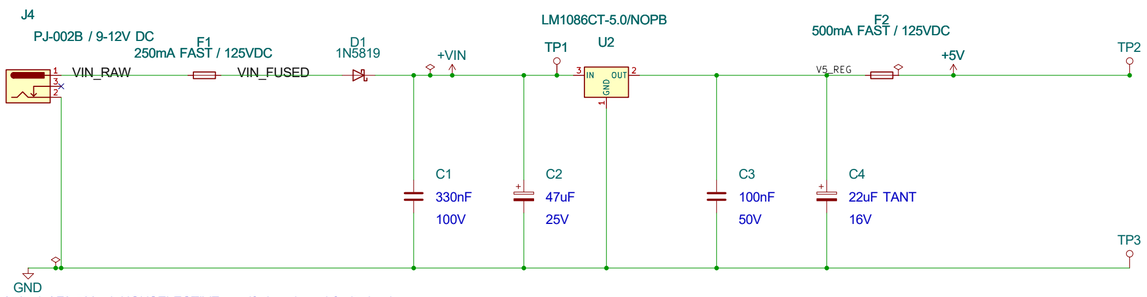
WEATHER GADGET — LCD DISPLAY SUBSYSTEM

Zander Boward | Team 202 | Revision L.1 | 9 November 2026

ONE A2 SHEET | 5 V PIC / LCD logic / backlight | Three UART links

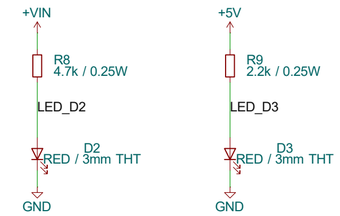
9–12 V INPUT / 5 V LINEAR REGULATOR

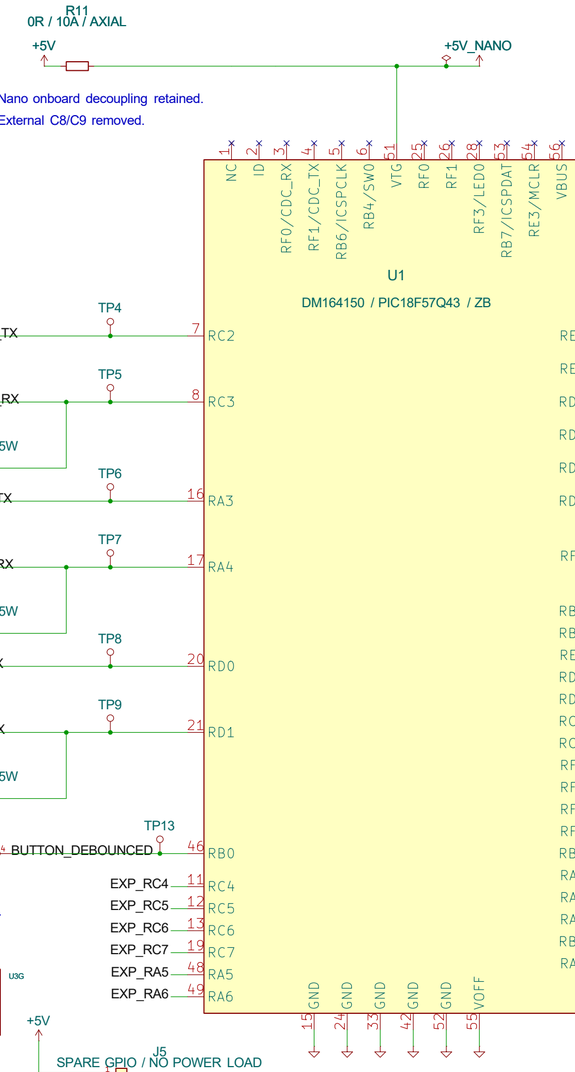


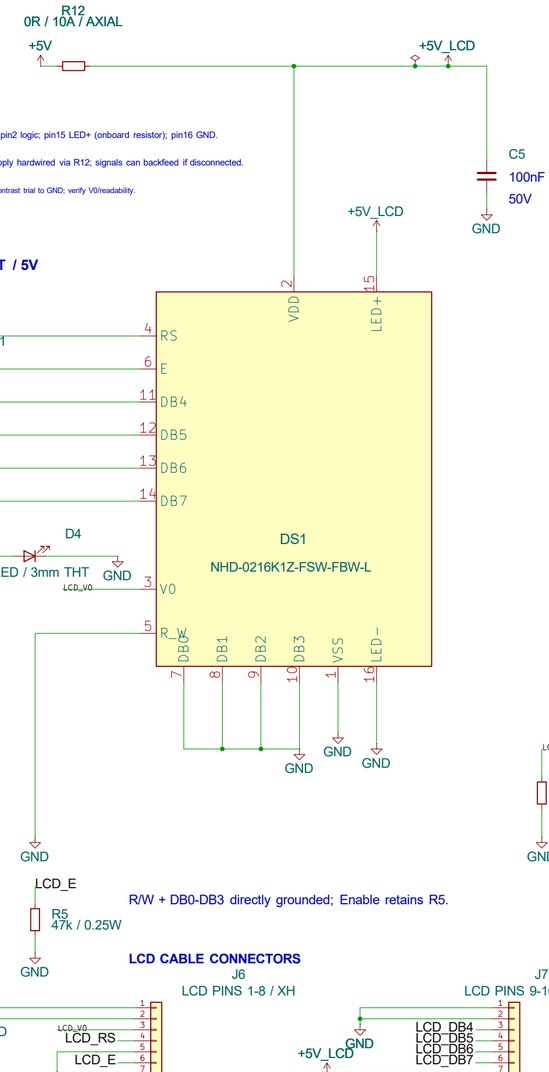
F1 250mA / F2 500mA NONSELECTIVE; verify inrush and fault clearing.

U2 LM1086: pin1=GND, pin2=OUT/TAB, pin3=IN. TAB IS LIVE 5V.

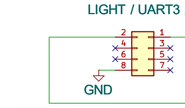
POWER INDICATORS





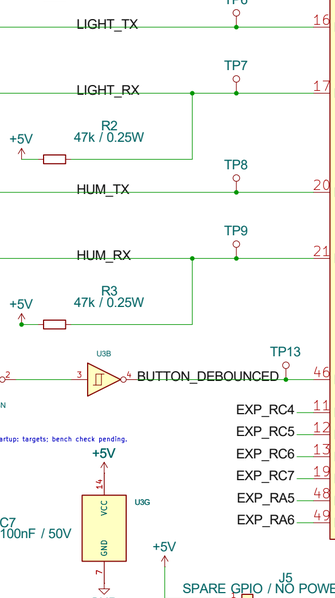


PRESSURE / UART1 / 5V LOGIC

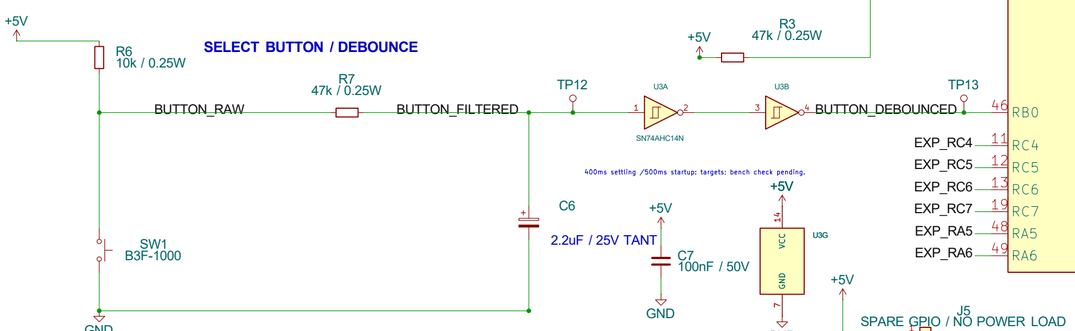


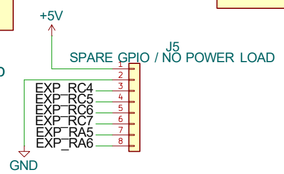
LIGHT / UART3 / 5V LOGIC

HUMIDITY / UART4 / 5V LOGIC

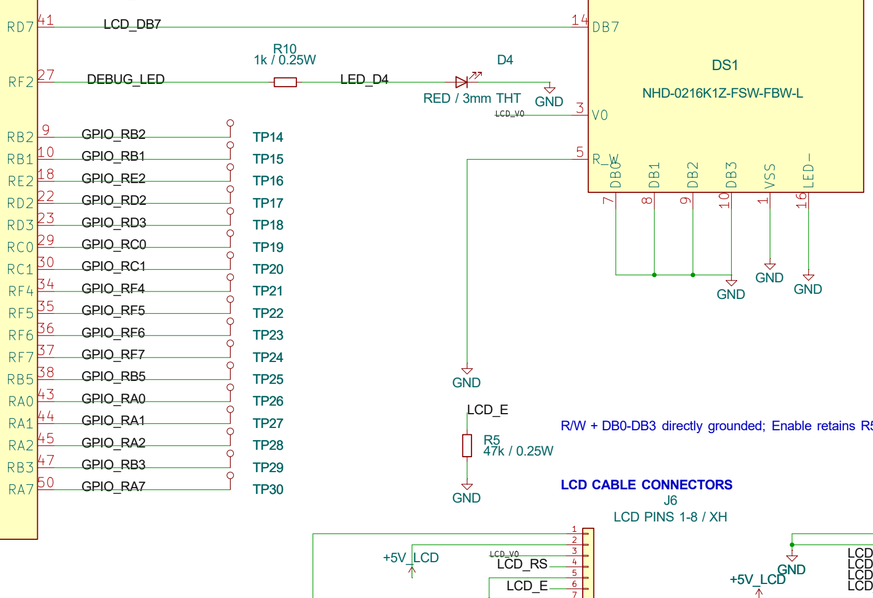


SELECT BUTTON / DEBOUNCE

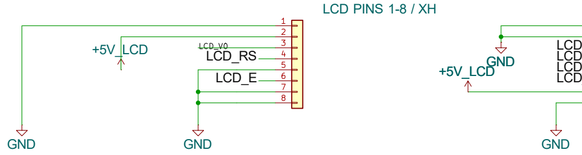


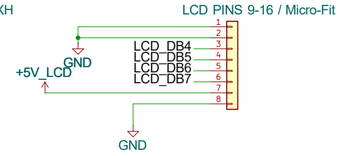


NEWHAVEN FBW-L / EXTERNAL CONTRAST / 5V



LCD CABLE CONNECTORS





16 LCD conductors. J6 XH and J7 Micro-Fit; solder display end at DS1.

UART: pin 1 local TX -> peer RX; pin 2 local RX <- peer TX; pin 8 GND; pins 3-7 NC. No power exported.

Verify peer 5 V compatibility and cable continuity. VTG 51 = 5 V; VOFF 55 = GND; VBUS 56 NC.

ALL SOURCES OFF before changing cables. Disconnect powered peers for unpowered tests.

100mA load envelope; Nano provisional. Bench voltage/thermal/inrush checks PENDING.

REVIEW DRAFT: hardware qualification pending; 0–40C; no J5 powered load
FBW-L LCD:AVX TAP C6;R/W and DB0–DB3 grounded;R13 contrast trial

Team 202 | Zander Boward

Sheet: /
File: Revision-L.kicad_sch

Title: Revision L.1 — PCB connector update

Size: A2	Date: 2026-11-09	Rev: L.1
KiCad E.D.A. 10.0.7		Id: 1/1